

Discussion on Bordenave, M., Role of forest biomass in the Finnish decarbonisation

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Overview

1. General comments
2. Major comments
3. Minor comments

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General comments

- The research question matters, predictions for Finland on the supply side of energy structure under energy transition, nuclear phase out and forest-biomass reservation.
- The research contribution is significant, employing typical day (TD) for end use energy demand and optimal energy supply portfolio, constructing a biomass supply curve.

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Major comments

- The introduction and literature review are concise and convincing.
- The results are presented beautifully in detail.
- The model description is not so clear by the structure and the content. It could be decomposed into three sections, stylized facts, model description, data specification and parameter calibration.
- The model description only shows 7 equations, which are not linked together. I cannot understand the model mechanism clearly but through guessing, especially section 3.3.2 end-use demand in energy services.
- It is not clear which part of the models are borrowed, which are innovated by the author.
- Exogenous end-use demand is a strong assumption, need more discussion as a limitation. Introducing additional scenarios, high/low energy demand in 2035, distributions?

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Minor comments I

- Page 7, add footnote the year 2017 was chosen as it is the historical year with the most complete data (i.e. the lowest number of hours with missing data, Thiran (2024)[p.60]).
- I suggest to put Figure 1-4 and their description in a stylized facts section before the model description section.
- Page 13, what is grid losses, $Loss(Elec, t)$?
- Page 15, equation (3), the cost minimization problem is minimizing with respect to what, to the installed size (Limpens et al., 2019)? The author needs to show the cost functions as well.
- Page 13 and 15, Moret (2017) and Thiran (2024) are PhD thesis, better to indicate the page as well.
- Page 15, the first two paragraphs of section 3.3.5 can be moved to the literature review section.

Minor comments II

- Page 17, the description of Table 4 is a bit messy, can be put below the table starting with "Note: $\cdots$".
- Page 19, how credible it is when the calibration reference year is also the initial period of the model? How many variables are set with the initial values among the total number of variables? What is the calibration method? Which parameters and variables are chosen for calibration? And what are their values, do they make sense?
- Page 22, Figure 10, what are the causes and consequences of this huge decrease in primary energy? I do not find imports (FI-5).
- Page 25, the references should be arranged by author surname in alphabetical order.

References I

Gauthier Limpens, Stefano Moret, Hervé Jeanmart, and Francois Maréchal. Energyscope td: A novel open-source model for regional energy systems. *Applied Energy*, 255: 113729, 2019.

Stefano Moret. *Strategic energy planning under uncertainty*. PhD thesis, Epfl, 2017.

Paolo Thiran. *Exploring options for a fossil-free European energy system: The role of renewable fuels*. PhD thesis, UCLouvain, 2024.

Good job and congratulations!